

Assignment

Answer the following

1. What is the difference between Single Quoted String and Double Quoted String in Python?

In Python, there is **no functional difference** between single quotes (' ') and double quotes (" "). Both are used to create strings.

Example:

```
name = 'Python'
language = "Python"
```

```
print(name)
print(language)
```

Output:

```
Python
Python
```

Use Case:

- Use single quotes when the string contains double quotes.
- Use double quotes when the string contains single quotes.

Example:

```
print("It's a beautiful day")
print('He said, "Hello!"')
```

2. What is the difference between Immutable and Mutable Objects?

Immutable Objects

Cannot be changed after creation.

New object is created when modified.

Examples: int, float, str, tuple

Mutable Objects

Can be modified after creation.

Original object is modified.

Examples: list, set, dict

Example:

```
x = "Python"
# x[0] = "J" # Error (immutable)
lst = [1, 2, 3]
lst.append(4)
print(lst)
```

Output:

```
[1, 2, 3, 4]
```

3. What is the difference between List and Tuple in Python?

List

Mutable

Uses square brackets []

Can be modified

Slower than tuple

Tuple

Immutable

Uses parentheses ()

Cannot be modified

Faster than list

Example:

```
my_list = [1, 2, 3]
my_tuple = (1, 2, 3)
my_list.append(4)
print(my_list)
```

Output:

```
[1, 2, 3, 4]
```

4. What are the differences between a Set and List in terms of Functionality and Use Cases?

List	Set
Ordered collection	Unordered collection
Allows duplicate values	Does not allow duplicates
Indexed	Not indexed
Uses []	Uses {}

Example:

```
my_list = [1, 2, 2, 3]
my_set = {1, 2, 2, 3}
```

```
print(my_list)
print(my_set)
```

Output:

```
[1, 2, 2, 3]
{1, 2, 3}
```

Use Cases:

- **List:** Storing ordered data, maintaining duplicates.
- **Set:** Removing duplicates, membership testing, set operations.

5. How does a Dictionary differ from a List in terms of Data Storage and Retrieval?

Dictionary	List
Stores data as key-value pairs	Stores data as ordered elements
Accessed using keys	Accessed using indexes
Faster lookup using keys	Lookup by value is slower
Uses {key: value} format	Uses []

Example:

List:

```
students = ["Rahul", "Amit", "Priya"]
print(students[1])
```

Output:

```
Amit
```

Dictionary:

```
student = {
    "name": "Rahul",
    "age": 20
}
```

```
print(student["name"])
```

Output:

```
Rahul
```